Aligning Metabolomic Networks via Graph Learning for Common Subnetwork Discovery

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Problem formulation

The mass spectrometry analysis includes two approach: **targeted** analysis and **untargeted** analysis.

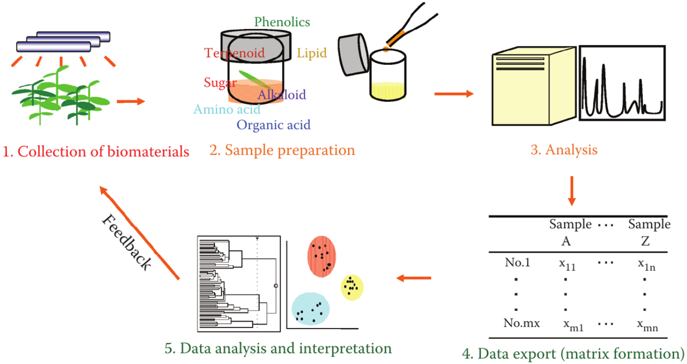

Figure 1: Metabolomics workflow.
Source: [1]

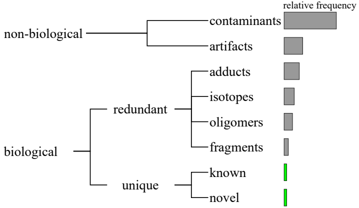

Figure 2: Untargeted composition.
Source: Adapted from [2]

Our objectives

Aligning metabolomic networks for filter Mass spectrometry data via Graph Neural Network models.

- Develop a pipeline for the analysis of metabolomic networks.
- Model metabolomic networks from mass spectrometry data.
- Implement GNN models to learn metabolite representations.
- Evaluate the effectiveness of the proposed GNN-based alignment approach.

Graph Neural Network (GNN)

It is a function $f : (A, X) \rightarrow Z$ that maps each node $v \in V$ of a graph $G = (V, E)$ to a low-dimensional embedding $z_v \in \mathbb{R}^d$ $(d \ll |V|)$.

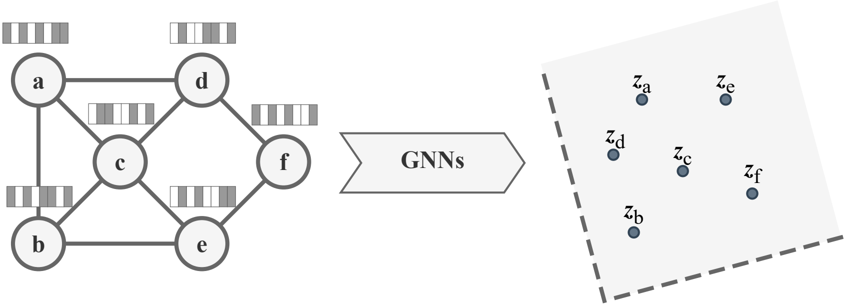

Figure 3: GNNs scheme.

Graph aligment problem

Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two graphs. The goal is to align their nodes through a mapping function $\pi : \mathbf{V_1} \to \mathbf{V_2}$, where each node $v \in V_1$ is mapped to its corresponding node $\pi(v) \in V_2$.

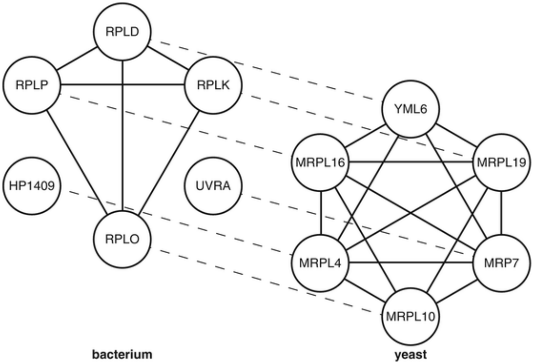

Figure 4: Node alignment.

Source: `https://link.springer.com/rwe/10.1007/978-1-4419-9863-7_992`

Base GNN model

T-GAE: Transferable Graph Autoencoder for Network Alignment [3].

Figure 5: T-GAE encoder.

Method

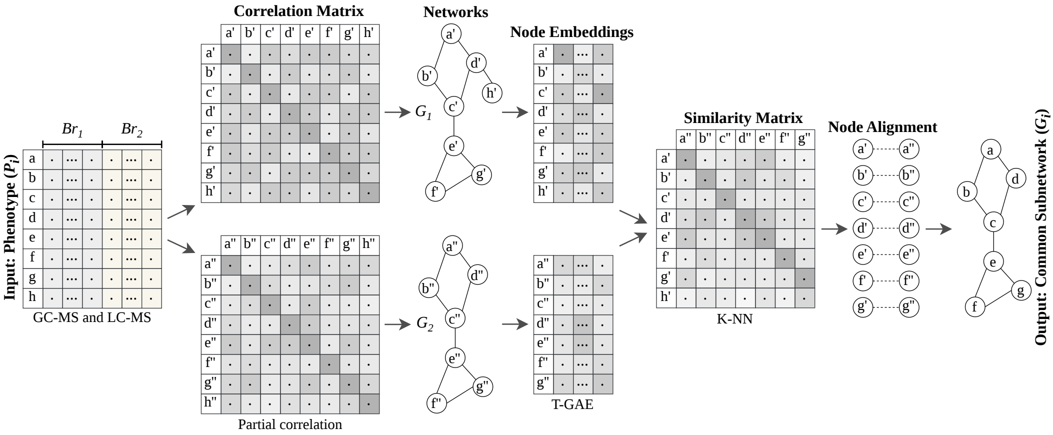

Figure 6: Our workflow.

Method overview

1. **Correlation estimation:** compute the partial correlation matrix $P = [\rho_{ij}]$ from mass espectrometry data.
2. **Network construction:** build a metabolic network using a correlation threshold ($|\rho_{ij}| \geq 0.5$), incorporate node features (like Rt, Mz, $Avg\ intensities$) and edge features (like $|\rho_{ij}|$, $\mathrm{sign}(\rho_{ij})$).
3. **Graph representation learning:** learn node embeddings with the `T-GAE` architecture based on `GIN` and `GINE` layers.
4. **Network alignment:** identify metabolite correspondences using a `KNN`-based similarity matrix.

Hardware/software

The experiments were run on an AMD Ryzen Threadripper PRO 5955WX computer with 128 cores, 256 GB of RAM, 2x NVIDIA RTX A6000 with 48 GB of VRAM. The source code was implemented in Python with PyTorch Geometric.

Dataset

These 4 datasets were used in the experiments.

- Synthetic
- Leaf [4]
- Mentos (from ICOBA-PUCP)
- Cacao (from ICOBA-PUCP)

Experiments setup

Table 1: Experiments setup.

Features	Values
Node embedding	`T-GAE`
Architectures	`GIN`, `GINE`
Dimensions	16, 32, 64
Optimizer	Adam with $lr = 10^{-4}$, $weight\_decay = 5 \times 10^{-4}$
Early stopping	$patience = 10$
Epochs	100
Node features	Mz, Rt, $\log(Avg\ intensities)$, $Std\ intensities$, $Cv\ intensities$
Edge features	$\lvert\rho_{ij}\rvert$, $\mathrm{sign}(\rho_{ij})$, ρ_{ij}^2
Dataset	Cacao, Leaf, Mentos

Build metabolomic networks

Table 2: Cacao dataset.

Group	Subgroup	$\|V\|$	$\|E\|$	Density	Is connected?
DryAmazonas	1	162	5001	0.383483	True
DryAmazonas	2	162	4096	0.314010	True
DrySanMartin	1	162	3817	0.292692	True
DrySanMartin	2	162	3728	0.285868	True
DryCusco	1	162	4265	0.327045	True
DryCusco	2	162	3348	0.256729	True
FreshAmazonas	1	162	3749	0.287478	True
FreshAmazonas	2	162	3946	0.302584	False
FreshSanMartin	1	162	3818	0.292769	True
FreshSanMartin	2	162	3460	0.265317	True
FreshCusco	1	162	3600	0.276052	True
FreshCusco	2	162	3669	0.281343	True

Node embedding (T-GAE)

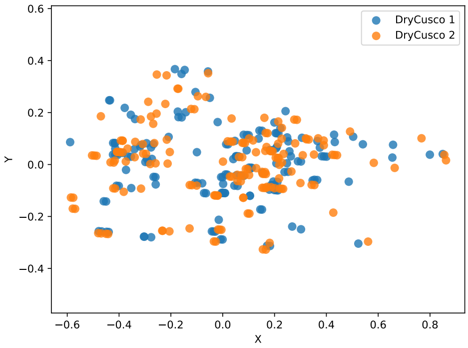

Figure 7: Node embeddings (DryCusco).

Network alignment

Table 3: Node aligment (DryCusco).

ID 1	ID 2
0	93
2	2
4	57
5	5
7	93
...	...
542	542
551	551
556	509
557	550
558	530

Table 4: Node filtered **294/559** (DryCusco).

ID	Rt	Mz	Metabolite name
2	0.097	172.95681	Unknown
5	0.201	144.98254	Unknown
15	1.871	118.08675	Betaine
30	1.949	446.88860	Unknown
46	2.110	118.08671	Betaine
531	29.792	374.30350	Docosahexaenoic acid ethyl ester
532	29.794	352.32167	Unknown
537	29.838	454.42642	N-Hexacosanoylglycine
538	29.839	150.12816	2,6-Diethylaniline
542	29.860	398.23344	Drotaverine
551	29.954	150.12807	2,6-Diethylaniline

Clustering analysis

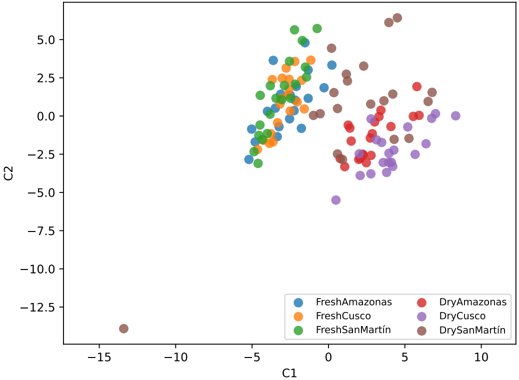

Figure 8: Clustering (DryCusco).

Comparison of encoders

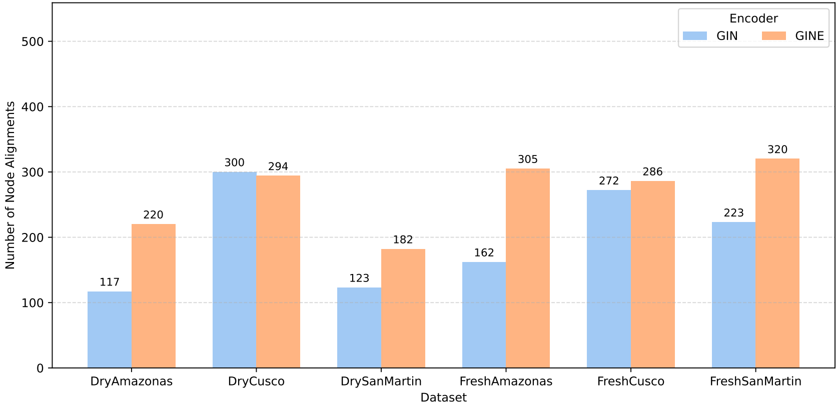

Figure 9: GIN vs. GINE.

Conclusions

Our proposal, is a approach to the exploratory analysis of metabolic networks.

- Features shared by cacao beans within the same group are retained, reducing the number of features before identification is attempted.
- A core graph embedding is generated for each cacao bean, providing a compact representation that can be used as input to a classifier for subsequent cacao bean identification.
- The alignment workflow enables samples acquired on different days to be integrated into a single dataset. This approach mitigates known batch effects, such as retention-time shifts, while the graph embedding favors correlations among signals rather than depending on their absolute intensity values.

Thanks!

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https://github.com/win7

[1] S.P. Putri and E. Fukusaki.
Mass Spectrometry-Based Metabolomics: A Practical Guide.
Taylor & Francis, 2014.

[2] Miriam Sindelar and Gary J Patti.
Chemical discovery in the era of metabolomics.
Journal of the American Chemical Society, 142(20):9097–9105, 2020.

[3] Jiashu He, Charilaos Kanatsoulis, and Alejandro Ribeiro.
T-gae: Transferable graph autoencoder for network alignment.
In *The Third Learning on Graphs Conference*, 2024.

[4] Marcia González-Teuber, Valeria Palma-Onetto, Carolina Aguirre, Alfredo J Ibáñez, and Axel Mithöfer.
Climate change-related warming-induced shifts in leaf chemical traits favor nutrition of the specialist herbivore battus polydamas archidamas.
Frontiers in Ecology and Evolution, 11:1152489, 2023.